

AMAZON SALES DATA ANALYSIS REPORT

NAME : Anusha G

ROLE : Data Analysis

TOOLS USED : Excel, SQL

INTRODUCTION

This project focuses on analysing Amazon sales data to identify key business insights such as sales trends, customer behaviours, product performance, and operational efficiency. The analysis was performed using SQL for data extraction and Excel for visualization.

OBJECTIVE

- To analysed overall sales performance
- To identify top-performing regions and categories
- To study customer buying patterns
- To evaluate delivery and fulfilment efficiency
- To create an interactive dashboard for business insights

DATASET DESCRIPTION

The dataset contains 1,20,378 records with the following key fields:

- Order ID
- Date
- Order Status
- Fulfilment Type
- Category
- Size
- Quantity (Qty)
- Amount
- State & City
- B2B Indicator

DATA CLEANING (EXCEL)

- Removed unnecessary columns
- Handled missing values
- Converted date format
- Removed duplicates
- The Size column was standardized by using Find and Replace (e.g., 'XL' to 'X-Large'), and the State column was converted to uppercase using the UPPERCASE formula."

TOOLS USED

- Excel: Data cleaning, Pivot Tables, Dashboard
- SQL: Data extraction and analysis

SQL ANALYSIS

Key Performance Indicators (KPIs)

1.Total Sales

```
select sum(Amount) as Total_Sales from amazon_sales ;
```

Result Grid	
Total_Sales	
▶	6207916329

2.Total Order

```
select count('Order ID') as Total_Order from amazon_sales ;
```

Result Grid	
Total_Order	
▶	120378

3.Total Quantity

```
select sum(Qty) as Total_Quantity from amazon_sales ;
```

Result Grid	
Total_Quantity	
▶	108774

4. Average Order Value

```
select sum(Amount)/ count('Order ID') as Average_Order_Value  
from amazon_sales ;
```

Result Grid	Filter Rows:
Average_Order_Value	
51570.1900	

5.Delivery Success Rate

```
select (count(case when status = 'Shipped' then 1 end) * 100.0  
/count(*)) as Delivery_Success_Rate  
FROM amazon_sales;
```

Result Grid	Filter Rows:
Delivery_Success_Rate	
83.47123	

6.Delivery Cancelled Rate

```
select (count(case when status = 'Cancelled' then 1 end) * 100.0  
/count(*)) as Delivery_Cancelled_Rate  
FROM amazon_sales;
```

Result Grid	Filter Rows:
Delivery_Cancelled_Rate	
16.52877	

Charts

7.Sales by Category

```
select Category, sum(amount) as Total_sale from amazon_sales
group by Category
order by Total_sale desc;
```

Result Grid	Filter Rows:
Category	Total_sale
Set	4168013749
Western Dress	1437858663
kurta	476952188
Top	93352248
Ethnic Dress	22479624
Bottom	4632355
Blouse	3208680
Saree	1382717
Dupatta	36105

8.Top 5 States

```
select State, sum(amount) as total_sales from amazon_sales
group by State
order by total_sales desc limit 5;
```

Result Grid	Filter Rows:
State	total_sales
KERALA	1961243252
MAHARASHTRA	672989518
UTTAR PRADESH	455542362
KARNATAKA	362501715
DELHI	331695523

9.Order status

```
select Status, count(*) as total_order from amazon_sales
group by Status
order by total_order desc;
```

Result Grid	Filter Rows:
Status	total_order
Shipped	100481
Cancelled	19897

10.Size Analysis

```
SELECT size, SUM(Qty) AS total_quantity
FROM amazon_sales
GROUP BY size
ORDER BY total_quantity DESC;
```

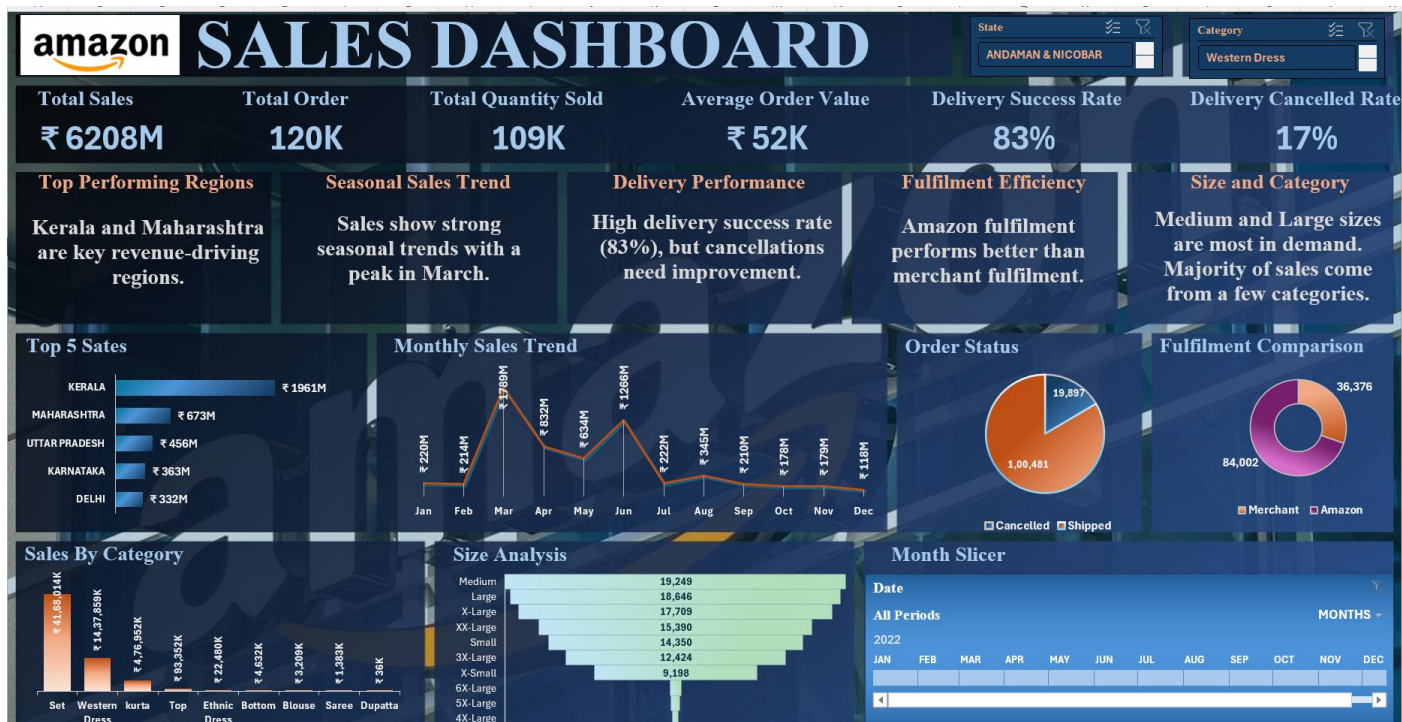
Result Grid			Filter Rows:
	size	total_quantity	
▶	Medium	19249	
	Large	18647	
	X-Large	17709	
	XX-Large	15390	
	Small	14350	
	3X-Large	12424	
	X-Small	9198	
	6X-Large	625	
	5X-Large	484	
	4X-Large	361	
	Free	337	

11.Fulfilment Comparison

```
select fulfilment, count(amount) as count_sales
from amazon_sales
group by fulfilment
order by count_sales ;
```

Result Grid			Filter Rows:
	fulfilment	count_sales	
▶	Merchant	36376	
	Amazon	84002	

DASHBOARD OVERVIEW



KEY INSIGHTS

Top 5 States

- Kerala is the highest revenue-generating state, contributing significantly more than other regions.
- Maharashtra and Uttar Pradesh also show strong performance.
- Sales are concentrated in a few key states, indicating regional dependency.

Monthly Sales Trend

- Sales show strong seasonality, with a clear peak in March.
- After March, there is a noticeable decline, followed by a smaller recovery in June.
- The later months show relatively stable but lower sales levels.

Order Status

- A large majority of orders are successfully shipped, indicating efficient order processing.
- However, a noticeable portion of orders are cancelled, which highlights an area for operational improvement.

Fulfilment Comparison

- Amazon fulfilment handles a significantly higher number of orders compared to merchant fulfilment.
- This suggests better efficiency and customer preference for Amazon-managed logistics.

Sales by Category

- A few categories such as Set and Western Dress contribute the majority of total sales.
- Other categories have relatively lower contributions, indicating uneven product performance.

Size Analysis

- Medium and Large sizes are the most frequently sold, showing strong customer preference.
- Extremely small and large sizes have lower demand, suggesting limited market need.

CONCLUSION

The analysis reveals that sales are influenced by seasonal trends, regional demand, and product preferences. Improving cancellation rates and focusing on high-demand sizes and categories can enhance business performance.

RECOMMENDATIONS

- Focus on high-performing states for marketing
- Increase stock for Medium & Large sizes
- Reduce order cancellations through better logistics
- Promote top-selling categories
- Improve merchant fulfilment efficiency